

ASESII 24A02 Final Project - Part 2

2-way ANOVA method

Group 02

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1 Abstract

2 Dataset selection

2.1 Data Overview

```
data <- read.csv("data/student_performance.csv", header = TRUE)

config <- list(
  file = "student_performance.csv",
  response = "final_exam_score",
  excluded = "final_exam_score"
)

data_raw <- read.csv(find_exam_data(config$file), check.names = FALSE)
predictors <- setdiff(names(data_raw), config$excluded)
analysis_data <- data_raw[, c(config$response, predictors), drop = FALSE]
```

```
if (anyNA(analysis_data)) stop("Declare a training-only missing-value plan.")
```

Table 1: Response and candidate predictors, with the integrity checks run before the split.

Variable	Role	Type	Distinct	Missing
student_id	Predictor	integer	1000	0
gender	Predictor	character	2	0
study_time_hours	Predictor	numeric	70	0
attendance_percent	Predictor	numeric	327	0
sleep_hours	Predictor	numeric	65	0
parental_education	Predictor	character	5	0
internet_access	Predictor	character	2	0
extracurricular_activities	Predictor	character	2	0
part_time_job	Predictor	character	2	0
previous_grade	Predictor	numeric	434	0
final_exam_score	Response	numeric	342	0
final_grade	Predictor	character	5	0

For 2-Way ANOVA, this method requires the dataset to have at least 2 categorical variables along with a continuous response variable for the analysis.

In general, from our chosen dataset, we decide to evaluate the factors that directly affect the student performance on the score. This data contains information over 1000 students, including the daily habits (the study time, hours of sleep, internet access), lifestyles (extracurricular activities, parttime job) demographic information (gender, education level of their parents) and academic performance (final exam score, final grade). It's ideal for exploratory data analysis and 2-Way ANOVA analysis based on the behavioral and demographic factors.

Specifically, there are 6 categorical factors: **gender** (Male/Female), **parental_education** (PhD/Master/Bachelors/School/None), **internet_access** (Yes/No), **extracurricular_activities** (Yes/No), **part_time_job** (Yes/No), **final_grade** (A/B/C/D/F). In this method, we pick the continuous response on **final_exam_score** to test the hypothesis whether any factors that affect this response as well as any relationships/interactions between 2 categorical factors.

2.2 Research Questions & Hypotheses

We investigate whether parental education and part-time job status are associated with students' final exam scores, and whether there is an interaction between the two factors

We follow the 2-way ANOVA model:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

with $\epsilon_{ijk} \sim N(0, \sigma^2)$.

- μ is the overall mean.
- α_i is the effect of the i -th level of Factor A, `parental_education`.
- β_j is the effect of the j -th level of Factor B, `part_time_job`.
- $(\alpha\beta)_{ij}$ is the interaction effect between Factor A and Factor B.
- ϵ_{ijk} is the random error term.

2.2.1 Factor A: Parental Education

Research Question:

Does participation in parental education affect students' final exam scores?

$$\begin{cases} H_{0a} : \alpha_1 = \alpha_2 = \dots = \alpha_5 = 0 \\ H_{1a} : \text{Not all } \alpha_i \text{ are equal to 0.} \end{cases}$$

2.2.2 Factor B: Part-time Job

Research Question:

Does having a part-time job affect students' final exam scores?

$$\begin{cases} H_{0b} : \beta_1 = \beta_2 = 0 \\ H_{1b} : \text{Not all } \beta_j \text{ are equal to 0.} \end{cases}$$

2.2.3 Interaction Effect

Research Question:

Does the effect of students' part-time job on final exam scores depend on their parental education?

$$\begin{cases} H_{0ab} : (\alpha\beta)_{ij} = 0 \quad \text{for all } i, j \\ H_{1ab} : \text{Not all } (\alpha\beta)_{ij} \text{ are equal to 0.} \end{cases}$$

2.3 Descriptive Statistics

2.3.1 Data Summary Statistics

Table 2: Overall Summary Statistics of Final Exam Score

N	Mean	Median	Variance	SD	Min	Max
1000	83.54	83.8	106.94	10.34	46.8	100

The dataset contains 1,000 observations of final exam scores. The mean score is 83.54, with a median of 83.80, indicating that the scores are centered around the mid-80s. The standard deviation is 10.34, showing moderate variability, while the scores range from 46.8 to 100. The relatively close mean and median suggest no strong skewness in the overall response distribution.

Table 3: Summary Statistics by Treatment Combination

Parental Education	Part-time Job	N	Mean	Median	Variance	SD	Min	Max
Bachelors	No	221	84.47	85.00	88.80	9.42	57.4	100
Bachelors	Yes	87	80.69	81.20	127.75	11.30	57.3	100
High School	No	238	84.53	84.45	102.76	10.14	54.1	100
High School	Yes	118	80.61	81.95	117.06	10.82	53.2	100
Masters	No	121	84.47	85.00	98.81	9.94	63.7	100
Masters	Yes	63	83.21	84.90	115.06	10.73	46.8	100
None	No	69	83.89	83.80	111.28	10.55	59.3	100
None	Yes	33	82.49	80.80	100.53	10.03	68.0	100
PhD	No	35	87.44	88.70	96.48	9.82	69.5	100
PhD	Yes	15	79.51	79.00	152.49	12.35	58.8	100

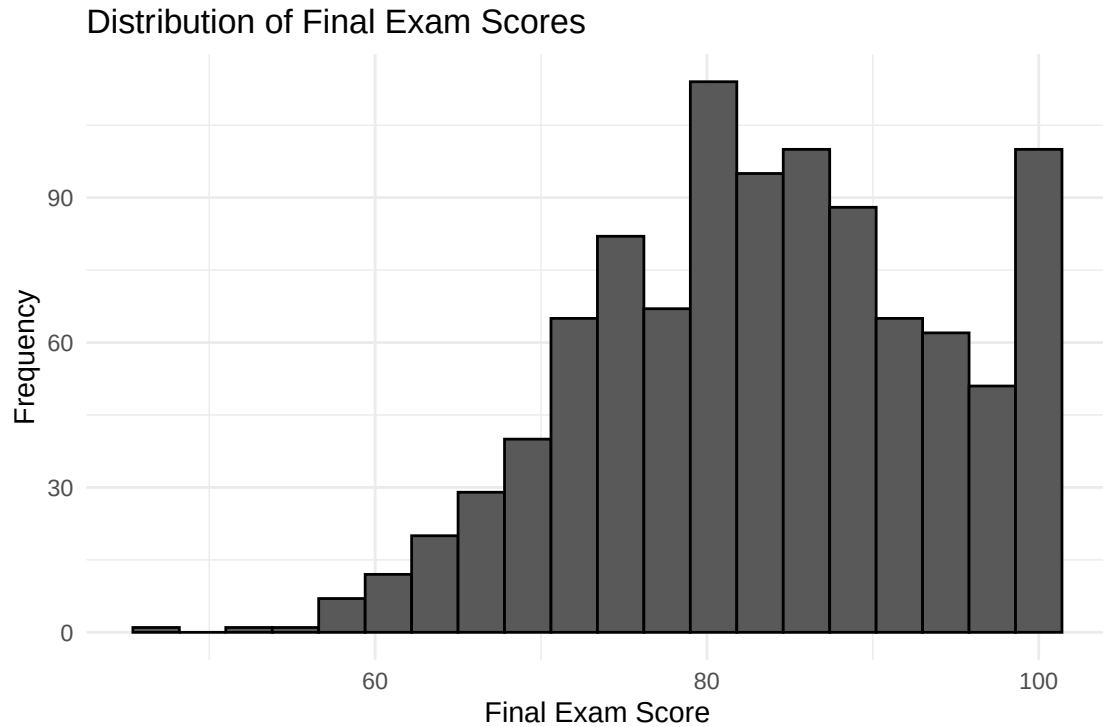
The dataset contains five levels of parental education and two levels of part-time job status. Students without a part-time job generally have higher mean final exam scores across all parental education levels.

The difference between students with and without a part-time job varies across parental education groups. The decrease is relatively small for students with Master’s or no parental education, while the largest decrease is observed among students with PhD-educated parents, from 87.44 to 79.51 points.

This variation suggests a potential interaction between parental education and part-time job status. The variability also differs across groups, with the PhD and part-time-job group having the largest variance (152.49). Therefore, homogeneity of variance should be carefully checked before conducting the two-way ANOVA.

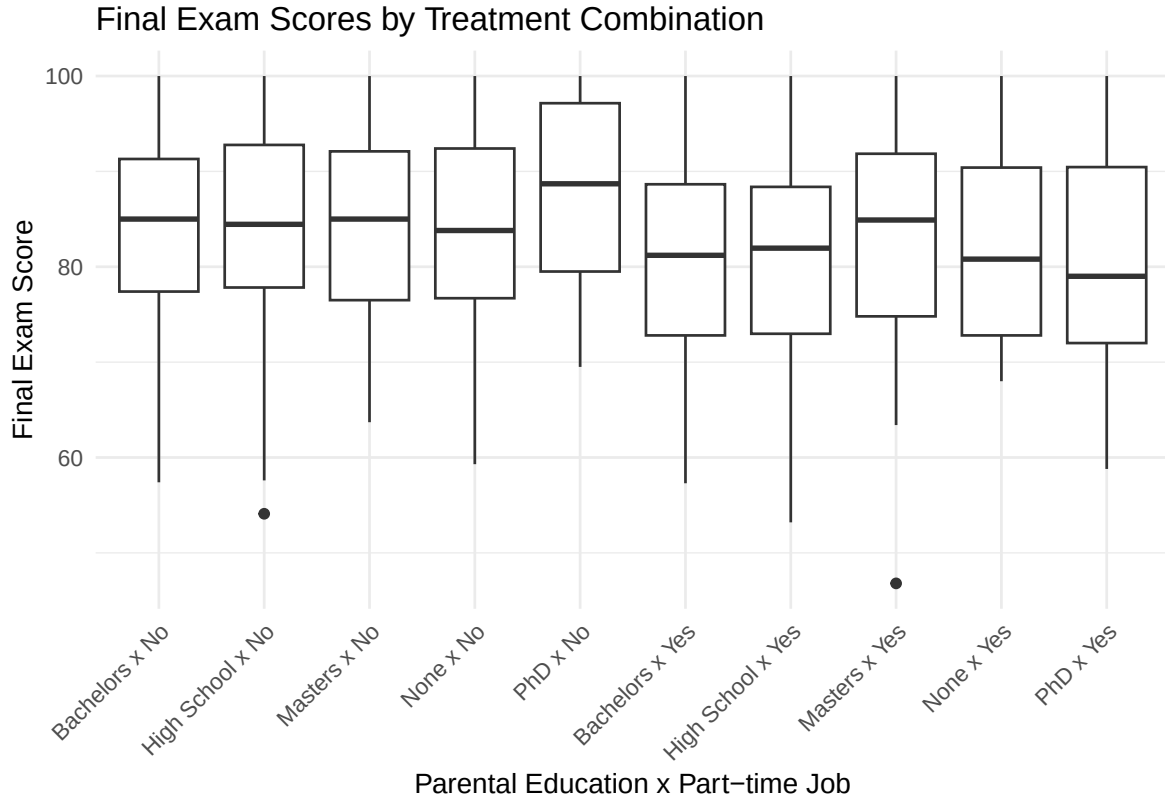
2.3.2 Visualization

2.3.2.1 Histogram of final score



The histogram shows that final exam scores range from approximately 47 to 100, with most observations concentrated between 70 and 100. The distribution shows a mild left skew, with a longer tail toward lower scores. However, the mean (83.54) and median (83.80) are close, suggesting that the skewness is not severe. A relatively large number of observations are also concentrated near the upper end of the score range, particularly around 95 to 100.

2.3.2.2 Box Plot

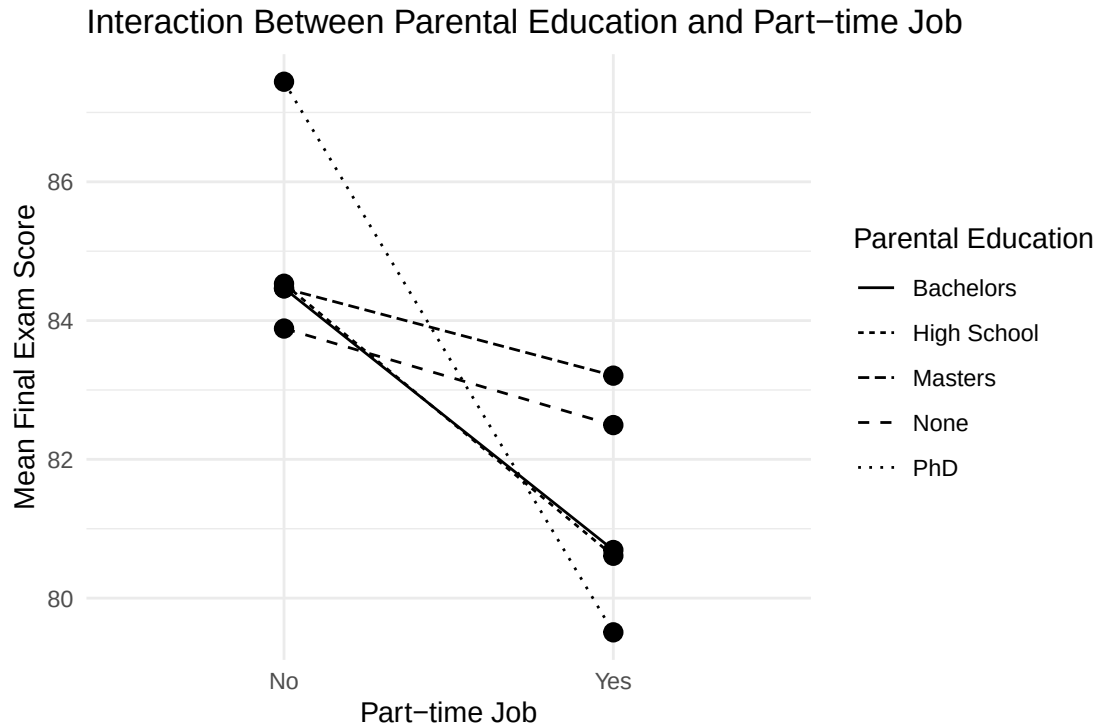


The box plot compares final exam scores across the ten treatment combinations of parental education and part-time job status. Students without a part-time job generally have higher median and mean final exam scores than students with a part-time job within the same parental education level.

The difference is particularly pronounced for students with PhD-educated parents, whose mean score decreases from 87.44 without a part-time job to 79.51 with a part-time job. In contrast, the differences between parental education groups are relatively small among students without a part-time job.

The spread of the groups is not completely uniform, with the PhD and part-time job group showing the largest variance. A few potential low outliers are also visible. Overall, the plot suggests that both parental education and part-time job status may be associated with final exam scores, while the different patterns across parental education levels suggest a possible interaction between the two factors.

2.3.2.3 Interaction Plot



The interaction plot shows the mean final exam score for each combination of parental education and part-time job status. Mean final exam scores decrease from the “No” to the “Yes” part-time job group across all parental education levels.

However, the magnitude of this decrease varies considerably across parental education levels. The mean score decreases by 3.78 points for students with Bachelor’s-educated parents and 3.92 points for students with High School-educated parents. In comparison, the decrease is only 1.26 points for students with Master’s-educated parents and 1.40 points for students with no parental education. The largest decrease is observed among students with PhD-educated parents, from 87.44 to 79.51 points.

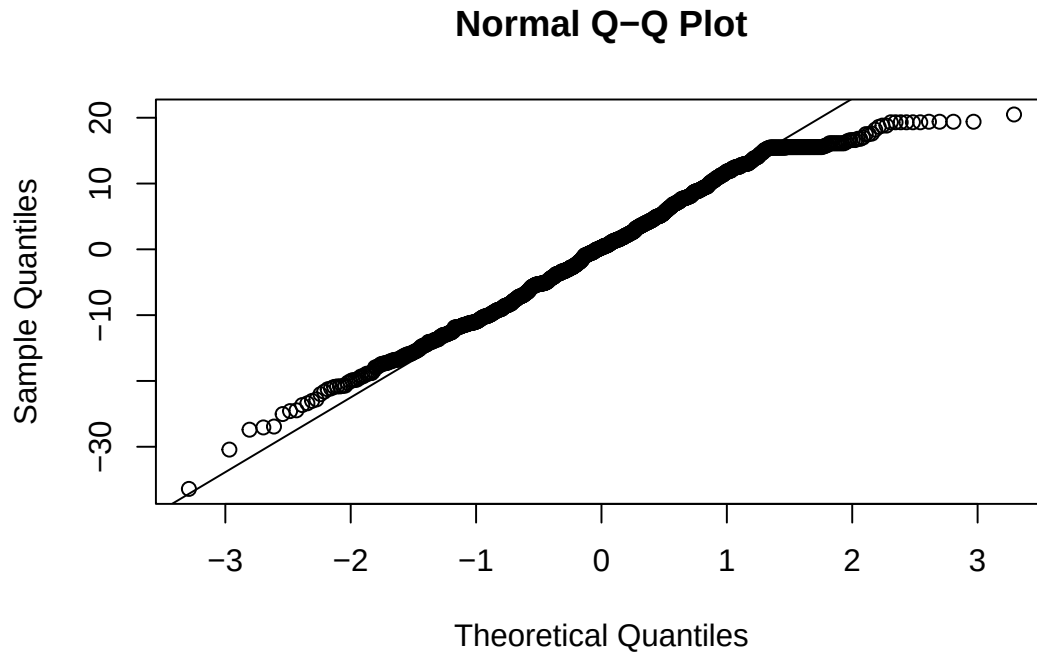
The lines are not parallel and several lines cross, suggesting a potential interaction between parental education and part-time job status. In particular, students with PhD-educated parents have the highest mean score when they do not have a part-time job but the lowest mean score when they have a part-time job.

This plot provides descriptive evidence of an interaction. Its statistical significance will be formally assessed using the two-way ANOVA.

2.3.3 Assumptions Check

```
qqnorm(residual)
qqline(residual)
```

2.3.3.1 Normality of Residuals



```
shapiro.test(residual)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residual
## W = 0.98486, p-value = 1.145e-08
```

The normality assumption was assessed using the normal Q-Q plot and the Shapiro-Wilk test. The hypotheses for the Shapiro-Wilk test are:

H_0 : The residuals follow a normal distribution

H_1 : The residuals do not follow a normal distribution

The Shapiro-Wilk test resulted in

$$W = 0.98486, \quad p = 1.145 \times 10^{-8}.$$

Since the p-value is less than 0.05, we reject H_0 . Therefore, there is statistical evidence that the ANOVA residuals deviate from a normal distribution.

The Q-Q plot also shows deviations from the reference line, particularly in the tails. However, the dataset contains 1,000 observations. With a large sample size, the Shapiro-Wilk test can be

highly sensitive to small departures from normality. Therefore, the normality assumption was not evaluated based on the p-value alone, and the Q-Q plot was also considered.

Overall, the residuals show some departure from normality, but the deviation should be considered together with the large sample size when evaluating the robustness of the ANOVA.

```
leveneTest(
  final_exam_score ~
    parental_education * part_time_job,
  data = data
)
```

2.3.3.2 Homogeneity of Variances

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value Pr(>F)
## group      9  0.6942 0.7147
##           990
```

The homogeneity of variance assumption was assessed using the treatment group box plots and Levene's test. The ten treatment combinations are formed by `parental_education` and `part_time_job`.

The hypotheses are:

$$H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2$$

$$H_1 : \text{At least one treatment group has a different variance}$$

Levene's test produced

$$F = 0.6942, \quad p = 0.7147$$

Since the p-value is greater than 0.05, we fail to reject H_0 . Therefore, there is no statistical evidence that the variances differ across the ten treatment combinations.

The group variances range from approximately 88.40 to 152.49, with the `PhD.Yes` group having the largest variance and the `Bachelors.No` group having the smallest variance. Although the group variances are not identical, Levene's test suggests that the differences are not statistically significant. Therefore, the homogeneity of variance assumption is considered satisfied for the subsequent two-way ANOVA.

```
sum(duplicated(data$student_id))
```

2.3.3.3 Independence of Errors

```
## [1] 0
```

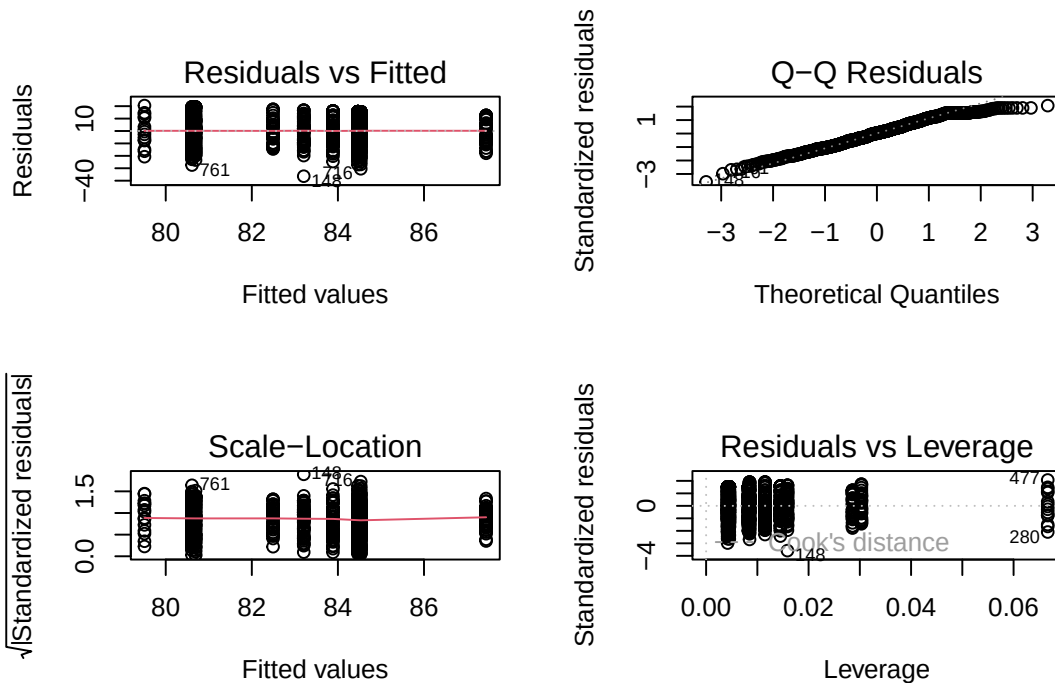
The independence assumption requires that the error terms are independent:

$$\epsilon_{ijk} \stackrel{ind}{\sim} N(0, \sigma^2).$$

Each observation corresponds to a different student, and no duplicated student IDs were identified. Therefore, the error terms are assumed to be independent. However, this assumption ultimately depends on the data collection process and cannot be formally verified from the dataset alone.

```
par(mfrow = c(2, 2))
plot(model)
```

2.3.3.4 Residual diagnostic plots



```
par(mfrow = c(1, 1))
```

The residual diagnostic plots show some deviation from normality in the Q-Q plot, particularly in the lower tail. The Residuals vs Fitted plot shows residuals distributed around zero without a clear systematic pattern. The Scale-Location plot also does not show a strong funnel-shaped pattern, which is consistent with the non-significant Levene's test ($p = 0.7147$).

The Residuals vs Leverage plot shows a few observations with relatively higher leverage, but there is no clear indication of a severely influential observation based on Cook's distance. Overall, the diagnostic plots suggest some deviation from normality, while the constant variance assumption appears reasonable.

2.3.4 Overall Assessment

The independence assumption is considered reasonable based on the structure of the dataset and the absence of duplicated student IDs. The normality assumption shows some deviation, particularly in the tails of the Q-Q plot, while the homogeneity of variance assumption is considered satisfied based on the non-significant Levene's test ($p = 0.7147$).

The normality deviation should be interpreted cautiously because the dataset contains 1,000 observations and the Shapiro-Wilk test is highly sensitive to small departures from normality.

Overall, the assumptions are considered reasonably adequate for conducting the two-way ANOVA. The observed deviation from normality should be acknowledged when interpreting the results, while no substantial concern is indicated for independence or homogeneity of variance.